

Machine Learning in Business

MIS710 – A2

Predicting Student Risk of Underperformance in Writing: A Machine Learning Approach



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Executive Summary

This report presents the findings of a comprehensive machine learning analysis performed for Data2Intel, an Australian learning analytics consulting service. The primary goal was to predict students at risk of underperforming in Year 3 writing assessments using a dataset collected from over 40 primary schools. The dataset contained detailed records on the students' reading and numeracy skills in Year 1 and Year 2, as well as key demographic information such as socioeconomic status (SES) and disability conditions.

To achieve this, we employed two supervised machine learning models: the Random Forest Classifier and the Gradient Boosting Classifier, with Gradient Boosting outperforming Random Forest in both accuracy and ROC AUC scores. The best model achieved an **accuracy of 73.67%** and an **ROC AUC of 0.7511**. In addition, we applied K-Means Clustering to identify potential student groupings based on their academic performance.

The report concludes by recommending Gradient Boosting as the most effective model for deployment, offering schools actionable insights to target at-risk students early. Furthermore, ethical considerations such as data privacy and bias are addressed, ensuring the responsible use of machine learning in education.

I. Introduction

Objective

Primary school students in Australia undergo yearly literacy and numeracy assessments, with Year 3 being a pivotal stage where students' foundational skills in reading and writing are evaluated through the National Assessment Program – Literacy and Numeracy (NAPLAN). Identifying students who are at risk of underperforming in writing early on can be instrumental in helping schools intervene before these students fall behind their peers. This project aims to develop predictive models that identify students at risk of underperformance based on their Year 1 and Year 2 academic records and demographic information.

Value Proposition

The ability to predict at-risk students allows schools to focus interventions, such as tutoring, specialized programs, or family engagement. Early intervention can improve student outcomes by addressing learning gaps early. For Data2Intel, this solution is part of a broader effort to enhance education standards across Australia through data-driven insights and targeted support.

2. Approach

Machine Learning Techniques

This project employed a combination of supervised learning and unsupervised learning to address the problem of predicting student underperformance.

Supervised Learning:

Two supervised machine learning models were developed to predict the binary target variable: "**Year3_Writing_At_Risk**," which indicates whether a student is at risk of failing Year 3 writing assessments based on their prior performance.

1. Random Forest Classifier:

The Random Forest algorithm is a robust and interpretable ensemble method that constructs multiple decision trees using different parts of the dataset and then averages the predictions. It is particularly effective for handling complex datasets with many features. The Random Forest model was chosen due to its ability to provide feature importance, helping educators understand which factors contribute most to student underperformance.

2. Gradient Boosting Classifier:

Gradient Boosting builds trees iteratively, focusing on the errors made by previous trees. This model works by minimizing a loss function and is highly effective for capturing non-linear relationships in the data. Gradient Boosting was chosen for its ability to provide higher accuracy in complex datasets, making it a more powerful predictor.

Unsupervised Learning:

Additionally, **K-Means Clustering** was used to explore natural groupings within the dataset. By clustering students based on their academic performance, the goal was to gain further insights into which groups of students might need targeted

interventions. This exploratory analysis allowed for the identification of distinct student profiles based on literacy and numeracy scores, as well as SES.

Prediction Target:

The prediction target, "Year3_Writing_At_Risk," was derived from students' NAPLAN writing scores in Year 3. Students classified as "at risk" have scores below the benchmark set by the Australian Curriculum, Assessment and Reporting Authority (ACARA), indicating a need for intervention.

3. Data Preparation and Exploratory Data Analysis (EDA)

Data Overview

The dataset comprises 2000 records, spanning over five years (2016–2020), and includes primary school students across 40 different schools in Australia. Each record contains several features, categorized into academic performance metrics (reading and numeracy scores), demographic details (socioeconomic status, family background), and disability conditions.

Key Features:

- Burt Reading Scores: A measure of students' literacy skills.
- Writing Vocabulary: Assesses the richness of students' writing.
- Numeracy Scores: Captures basic arithmetic skills such as counting and problem-solving.
- SES: Socioeconomic status, with a focus on students' family backgrounds.
- Disability Conditions: Any identified learning or physical disabilities that may impact academic performance.

Data Quality and Preprocessing

The dataset was clean, with no missing values, which facilitated a smoother preprocessing phase. However, data normalization was required to ensure that features such as reading and numeracy scores, which were measured on different scales, could be effectively used in the machine learning models.

Key preprocessing steps:

1. Standardization: Numerical features, including reading and numeracy scores, were standardized to have a mean of 0 and a standard deviation of 1.

Exploratory Data Analysis (EDA)

1. Socioeconomic Status (SES) Analysis:
 - Students' SES scores were examined across Year 1 and Year 2, revealing a significant correlation between low SES and being at risk of underperforming in writing. Students from lower SES backgrounds were disproportionately represented among those classified as "at risk" in the dataset. This finding underscores the importance of socioeconomic factors in early education.
2. Literacy and Numeracy Scores:
 - Violin plots were generated to visualize the distribution of literacy scores (such as Burt Reading and Writing Vocabulary) for both at-risk and not-at-risk students. The analysis showed that students flagged as "at risk" consistently scored lower in literacy tests compared to their peers who were not at risk.
3. Correlation Heatmap:
 - A correlation heatmap was created to investigate the relationships between academic scores and demographic factors. Strong correlations were observed between numeracy and literacy scores, with students who performed well in reading assessments also excelling in numeracy tests. This guided feature selection for the machine learning models, as these features were identified as key predictors.

4. Model Development and Evaluation

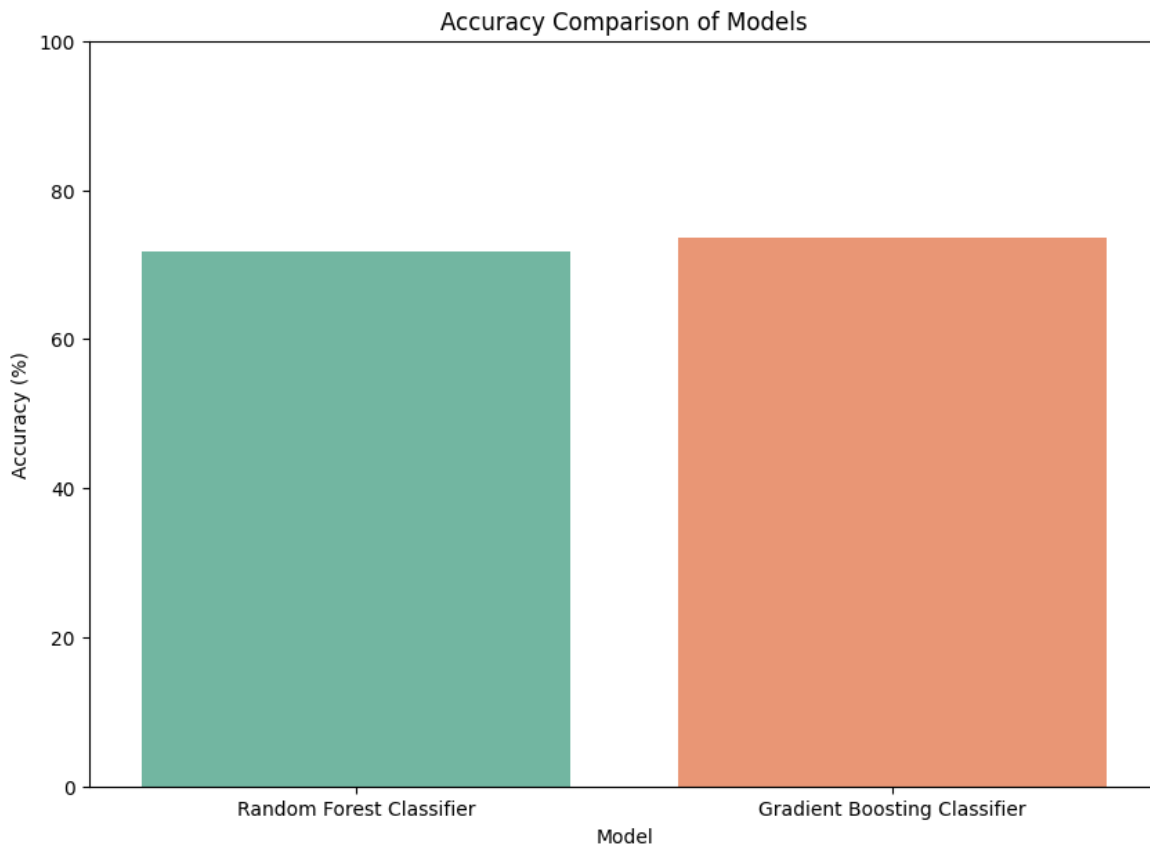
Supervised Learning Models

1. Random Forest Classifier:

- **Accuracy: 71.67%**
- **ROC AUC: 0.7431**
- The Random Forest model provided valuable insights into feature importance, indicating that literacy scores (such as Burt Reading and Writing Vocabulary) and socioeconomic status (SES) were the most important factors in predicting whether a student was at risk of underperforming. This model performed reasonably well in distinguishing between at-risk and not-at-risk students but was slightly less effective in predicting the minority class (students at risk).

2. Gradient Boosting Classifier:

- **Accuracy: 73.67%**
- **ROC AUC: 0.7511**
- Gradient Boosting achieved a higher accuracy than Random Forest and demonstrated a greater ability to capture the complex relationships between features in the dataset. The model performed well in identifying the subtle patterns that contribute to student underperformance, making it a better predictor overall.

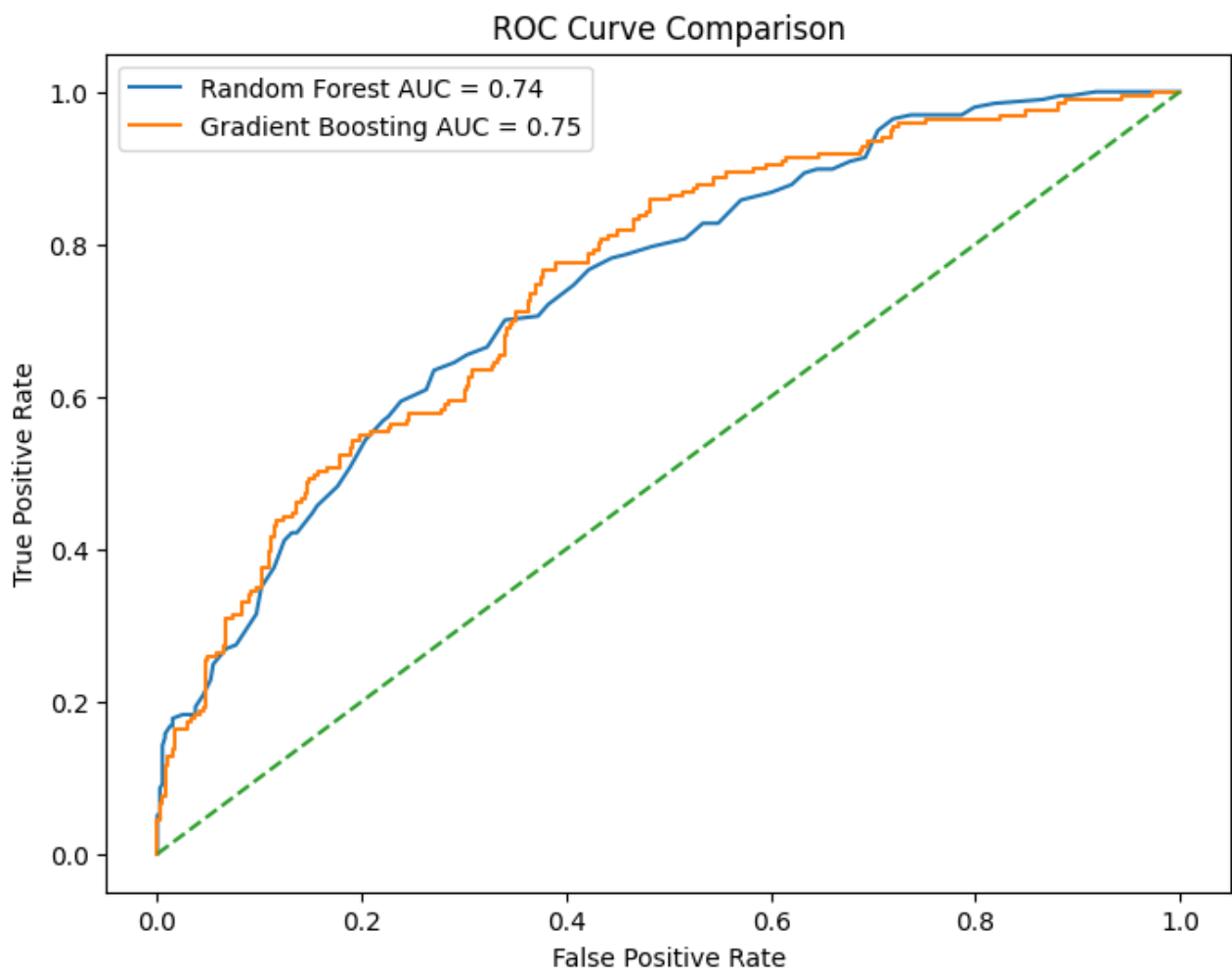
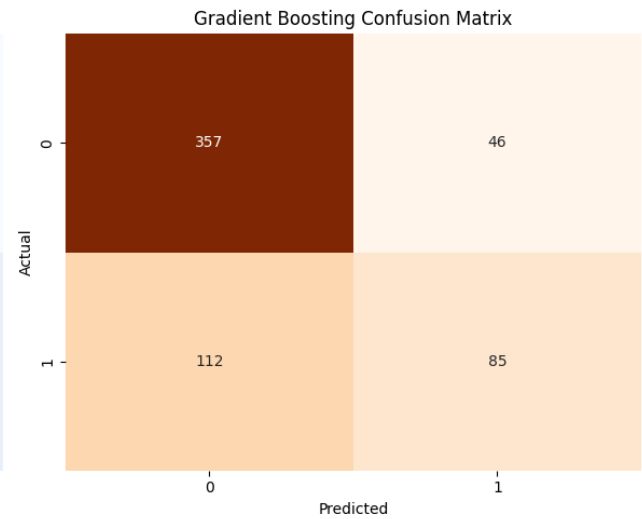
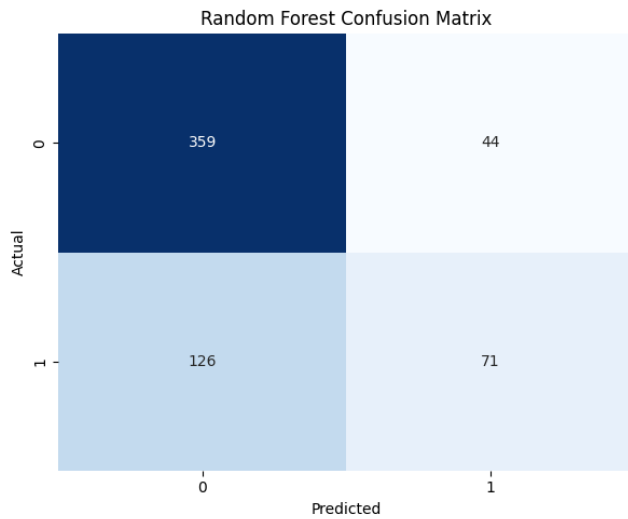


Model Comparison

The **Random Forest Classifier** achieved an accuracy of **71.67%** with an **ROC AUC** of **0.7431**. While it provided valuable insights into key predictors like literacy scores and SES, it struggled to accurately classify the minority class (students at risk). The **confusion matrix** revealed that although it performed well for the majority class, it had difficulty identifying some students who were indeed at risk.

In comparison, the **Gradient Boosting Classifier** outperformed Random Forest with a higher accuracy of **73.67%** and an **ROC AUC** of **0.7511**. It excelled at capturing complex relationships between features and showed better performance in reducing **false negatives**. This makes Gradient Boosting the more reliable model for

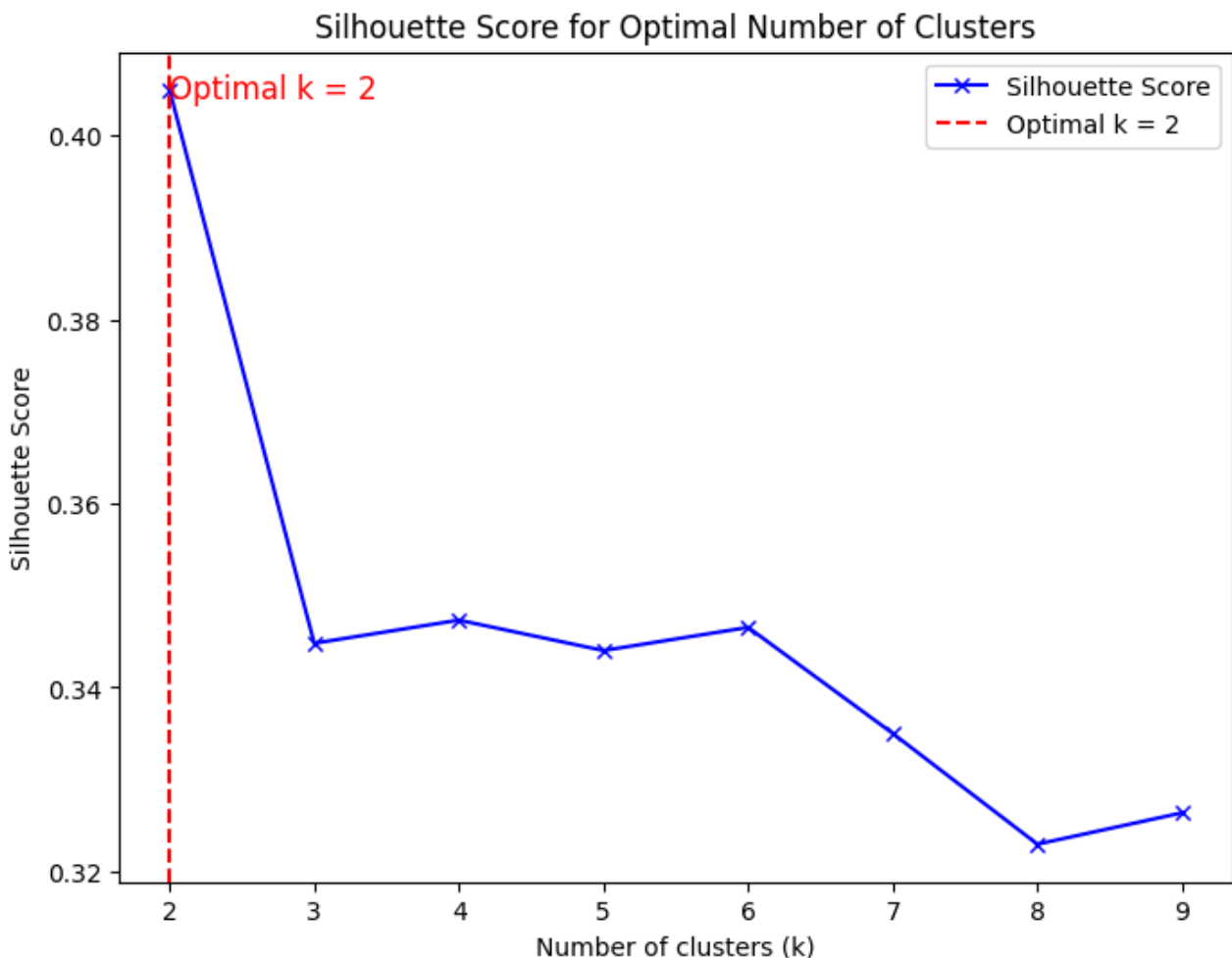
accurately predicting at-risk students.



Unsupervised Learning: K-Means Clustering

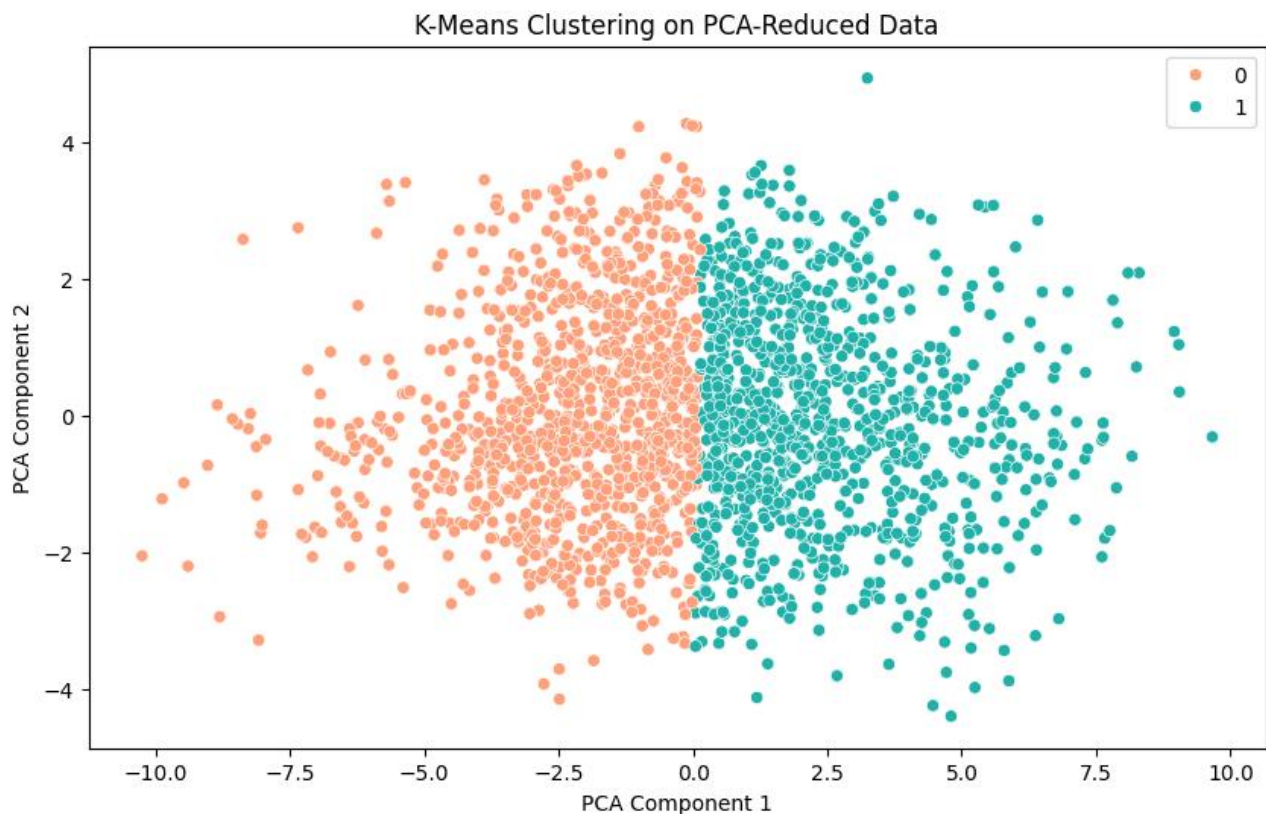
- K-Means Clustering was applied to explore potential groupings of students based on their academic performance. Using Principal Component Analysis (PCA) to reduce dimensionality, the optimal number of clusters was determined to be 2 based on the Silhouette Score. This analysis revealed two distinct clusters of students: (refer figure 6)
 - High-Performing Cluster: Students in this cluster had higher scores across literacy and numeracy metrics, and a higher SES on average.
 - Low-Performing Cluster: Students in this cluster tended to have lower literacy and numeracy scores and were more likely to be flagged as "at risk."

The clustering analysis provided additional insights into student profiles, helping to identify students who may need more intensive support, even if they are not currently flagged as "at risk."



5. Solution Recommendation

Model Recommendation:



Based on the model evaluations:

- Gradient Boosting Classifier is recommended for deployment due to its higher accuracy (73.67%) and better classification of students at risk, as evidenced by its higher ROC AUC (0.7511). The model's ability to capture complex relationships between the features makes it the ideal choice for predicting student underperformance.
- Random Forest Classifier can serve as a secondary model, offering greater interpretability. Its feature importance rankings provide educators with insights into which factors are most predictive of student success or failure, which can help guide future interventions.

Actionable Insights for Schools:

1. Targeted Interventions:

- Schools should focus their resources on students identified as "at risk" by the Gradient Boosting model. These interventions may include additional literacy tutoring, engagement with families to provide support at home, and tailored educational programs designed to address specific learning gaps.

2. Cluster-Based Support:

- The results from K-Means Clustering can further guide interventions by identifying student groups that share similar academic profiles. Students in the lower-performing cluster should be monitored closely and provided with additional support, even if they are not yet flagged as at risk.

Future Engagement:

To maintain the model's accuracy and relevance, it is recommended that Data2Intel work with the client (the consortium of primary schools) to periodically update the model as new data becomes available. By retraining the model with more recent student data, its predictive power can be improved. Additionally, incorporating new features, such as behavioural data or attendance records, could lead to even more accurate predictions.

6. Technical Recommendations

Development and Testing Environment:

- Programming Language: The entire project was developed using Python, with the code executed in Google Collab, which offers a cloud-based environment for running machine learning models.
- Libraries Used:
 - pandas: Data manipulation and preprocessing.
 - scikit-learn: Implementation of machine learning models and evaluation metrics.
 - matplotlib and seaborn: Data visualization for EDA and performance analysis.

- XGBoost: An implementation of Gradient Boosting for supervised learning.

Machine Learning Pipeline:

1. Data Preprocessing: Data was pre-processed by standardizing numeric variables and one-hot encoding categorical variables.
2. Model Training and Evaluation: Both Random Forest and Gradient Boosting models were trained using the pre-processed data, and their performances were evaluated using accuracy, ROC AUC, and confusion matrices.
3. Clustering: K-Means clustering was applied, and PCA was used to reduce dimensionality, enabling the identification of two distinct student clusters.

Deployment Plan:

For deployment, we recommend integrating the Gradient Boosting model into the school system's database, where it can be used to generate predictions based on new student data. The system should be updated periodically to incorporate new student records and ensure that the model remains relevant.

Maintenance of Accuracy:

To maintain the model's accuracy, it is essential to retrain the model regularly with new data. The addition of new features, such as family engagement and student behavioural data, could improve the model's ability to predict student underperformance. Additionally, a system for continuous performance monitoring should be established to detect any degradation in the model's accuracy over time.

8. References

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- **GradientBoostingClassifier**. (2024). Scikit-Learn. <https://scikit-learn.org/dev/modules/generated/sklearn.ensemble.GradientBoostingClassifier.html>
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